

CMP-20 - Constitutional Firewall

Enforce cross-domain identity, data, capability, authority, and evidence exchange against active signed Treaty decisions.

Authority and state ownership

Book I: Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58. Book II: Chapters 20, 21. Authoritative state: firewall decisions, cross-domain session state, Treaty enforcement cache, exchange audit bindings, cross-domain incident records. Explicitly non-owned: Treaty proposals or approval, native authority grants, external authoritative Identity state, payload source state.

Critical invariant

No external exchange bypasses the Constitutional Firewall, and a Treaty cannot authorize what Book I prohibits.

Logical interfaces

CMP-20-IF-01 / Command / Open Domain Session

Command `Open Domain Session` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-02 / Command / Authorize Exchange

Command `Authorize Exchange` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-03 / Command / Terminate Exchange

Command `Terminate Exchange` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-04 / Command / Quarantine Payload

Command `Quarantine Payload` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-05 / Query / Get Firewall Decision

Query `Get Firewall Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-06 / Query / Get Active Treaty View

Query `Get Active Treaty View` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-07 / Query / Explain Denial

Query `Explain Denial` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-08 / Event / Exchange Allowed

Event `Exchange Allowed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-09 / Event / Exchange Denied

Event `Exchange Denied` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-20-IF-10 / Event / Domain Session Terminated

Event `Domain Session Terminated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Closed > Negotiating > Active > Restricted > Closed. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: Treaty mismatch, identity substitution, policy-cache staleness, exfiltration attempt, domain compromise. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-20-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-20-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-20-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-20-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-20-TST-005 - Critical invariant. No external exchange bypasses the Constitutional Firewall, and a Treaty cannot authorize what Book I prohibits. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-20-TST-006 - Failure containment. Inject Treaty mismatch, identity substitution, policy-cache staleness, exfiltration attempt, domain compromise. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-20-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-20-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-20-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-20-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.